

PMNS Neutrino Mixing and the Solar-Isocurvature Cross-Sector Identity

David A. Besemer^{1,*}

¹*Independent Researcher*

(Dated: June 29, 2026)

We derive the PMNS neutrino mixing angles from the six-cone spatial partition of SDGFT. We establish the solar-isocurvature cross-sector identity, which couples the solar mixing angle $\sin^2 \theta_{12}$ to the cosmic neutrino isocurvature fraction $\beta_{\text{iso}} = 1/36$, yielding $\sin^2 \theta_{12} = 11/36 \approx 0.3056$. The atmospheric angle is shown to be $\sin^2 \theta_{23} = 41/72 \approx 0.5694$.

I. INTRODUCTION

Neutrino mixing angles are larger than quark mixing angles. We explain this difference through the direct projection of the spatial six-cone geometry.

II. THEORETICAL FORMULATION

Here we formulate the core mathematical relations governing the system:

$$\sin^2 \theta_{12} = \frac{1}{3} - \beta_{\text{iso}} = \frac{1}{3} - \frac{1}{36} = \frac{11}{36} \approx 0.30556 \quad (1)$$

$$\sin^2 \theta_{23} = \frac{41}{72} \approx 0.56944 \quad (2)$$

III. CROSS-SECTOR IDENTITY

The solar angle is coupled to the early-universe isocurvature fraction, demonstrating that flavor physics and cosmology are intimately linked.

IV. CONCLUSION

The PMNS angles are topologically locked, providing a testable prediction for neutrino oscillation experiments.

-
- [1] D. A. Besemer, “Axiomatic Foundations of SDGFT,” SDGFT-2026-01 (in preparation).
[2] D. A. Besemer, “Effective Spectral Dimension and the Banach Fixed-Point Theorem in SDGFT,” SDGFT-2026-04.

- [3] D. A. Besemer, `sdgft` Python package, <https://github.com/david-besemer/sdgft> (2026).

* david.besemer@sdgft.org